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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (Biotechnology) (CBCS) Degree Examinations

December 2023

BT2004 -1 – UNIT OPERATIONS IN BIOPROCESS INDUSTRIES

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following is a derived unit?
A) Force
B) Mass
C) Length
D) Time
2. Manometer is a device used to measure
A) Force
B) Length
C) Pressure
D) Mass
3. For a static fluid, how does the pressure vary?
A) Horizontally
B) With depth only
C) Both with depth and along horizontal direction
D) Neither with depth nor along horizontal direction
4. If the gauge pressure measured by manometer is 25000 Pa, then the absolute pressure is
A) 126325 Pa
B) 76325 Pa
C) 25000 Pa
D) 50000 Pa
5. Which of the following forces are considered in Euler's equation of motion?
A) Viscous and turbulent forces
B) Gravitational and compressibility forces
C) Pressure and gravitational forces
D) Pressure and viscous forces
6. Which of the following is not an assumption made in deriving Bernoulli's equation of motion
A) The flow is steady
B) The flow is rotational
C) The fluid is non-ideal
D) The flow is incompressible
7. For viscous flow through circular pipes, head loss due to friction is
A) Directly proportional to viscosity
B) Inversely proportional to viscosity
C) Directly proportional to the square of viscosity
D) Inversely proportional to the square of viscosity
8. The resistance to filtration is offered by
A) Pump, filter cake, and filter medium
B) Pump, filter medium, and filtrate
C) The feed, filter cake, and the filter medium
D) Slurry, filtrate, and cake
9. For constant pressure filtration, the graph of t/V v/s V gives an intercept of
A) $\Delta P/\mu^2$
B) $\mu R_m/\Delta P$
C) $\mu \alpha c/2A^2\Delta P$
D) $A/\Delta P$
10. When the fall of particles is not affected by the walls of the container and the neighboring particles, then such type of settling is called as
A) Free settling
B) Hindered settling
C) Dispersed settling
D) Forced settling
11. The two basic components of a centrifuge are
A) Motor and rotor
B) Mortar and pestle
C) Motor and shaft
D) Handle and motor
12. Isopycnic means
A) Equal density
B) Equal velocity
C) Equal viscosity
D) Equal pressure
13. Turbines are preferred for
A) High viscous fluids
B) Low viscous fluids
C) For sticky materials
D) For solids

14. Size reduction by rubbing gives
 A) Course products B) Very fine products
 C) Chunks D) Hard solids
15. 200-mesh screen will have how many openings per linear inch?
 A) 2 B) 20
 C) 200 D) 2000
16. Power number is a function of
 A) Impeller Reynold's and Froude's number B) Prandtl and Grashof number
 C) Prandtl and Froude's number D) Reynold and Grashof number
17. With respect to the diameter of the tubing, diameter of a pipe is
 A) Larger B) Smaller
 C) Equal D) Two times smaller
18. Which among the following is not a type of reciprocating pump?
 A) Centrifugal pump B) Piston pump
 C) Plunger pump D) Diaphragm pump
19. The ratio of throat diameter to orifice diameter is
 A) 0.25 B) 0.5
 C) 0.75 D) 1
20. The three parts of a venture meter are
 A) Throat, orifice, plug B) Pipe, orifice plate, vena contracta
 C) Suction pipe, piston, delivery pipe D) Converging cone, throat, diverging cone

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Define derived units. Give any two examples. | 3 | L*1 | 1 | 1 |
| b) Name the types of fluids based on Newton's Law of viscosity. | 3 | L1 | 1 | 1 |
| c) Water flows through a pipe AB 1.2 m diameter at 3 m/s and then passes through a pipe BC 1.5 m diameter. At C, the pipe branches. Branch CD is 0.8 m in diameter and carries one third of the flow in AB. The flow velocity in branch CE is 2.5 m/s. Find the volumetric flow rate in AB, the velocity in BC, the velocity in CD and the diameter of CE. | 10 | L3 | 2 | 2 |
| 2. a) State Pascal's Law of hydrostatic equilibrium for a static fluid. Derive the same. | 10 | L3 | 1 | 2 |
| b) Derive the shear stress distribution profile for laminar/viscous flow through circular pipes. | 6 | L3 | 2 | 2 |
| 3. a) A differential manometer is connected at the two points A and B of two pipes as shown in the Fig.3(a). The pipe A contains a liquid of specific gravity 1.5 while pipe B contains a liquid of specific gravity 0.9. The pressures at A and B are $9.81 \times 10^4 \text{ N/m}^2$ and $17.658 \times 10^4 \text{ N/m}^2$, respectively. Find the difference in mercury level in the differential manometer. | 6 | L3 | 1 | 2 |

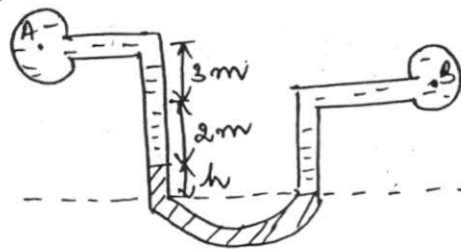


Fig.3(a).

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|--|---|----|---|---|
| b) Add a note on Reynold's experiment with a description on the Reynold's apparatus. | 5 | L2 | 2 | 1 |
| c) List any five reasons for minor energy losses during flow through pipes. | 5 | L1 | 2 | 1 |

Unit – II

- | | | | | | |
|-------|---|----|----|---|---|
| 4. a) | What is the difference between sedimentation and centrifugation? List any two types of sedimentation. | 4 | L1 | 3 | 1 |
| b) | Explain about plate and frame filter press. | 6 | L1 | 3 | 1 |
| c) | Describe the process of sieve analysis. | 6 | L2 | 4 | 1 |
| 5. a) | Differentiate between the following: i) Constant pressure and constant volume filtration ii) Batch and continuous filtration. | 6 | L2 | 3 | 1 |
| b) | What are the different laws governing the process of size reduction? Define each of them with mathematical expression. | 10 | L2 | 4 | 1 |
| 6. a) | What is the function of filter aids? List any two materials used as filter aids. | 4 | L1 | 3 | 1 |
| b) | List the assumptions of Kynch theory. | 4 | L1 | 3 | 1 |
| c) | Explain about any two types of agitators used in bioprocess industries. | 8 | L2 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Illustrate centrifugal pump. Explain its working mechanism. | 8 | L2 | 5 | 1 |
| b) | Explain the construction and working of venture meter. | 8 | L2 | 5 | 1 |
| 8. a) | Water is flowing at a rate of 500cm ³ /s through an orifice of 25mm diameter situated in 75mm diameter pipe. What will be the difference in the level on a mercury manometer connected across the meter? The co-efficient of meter is 0.65. | 8 | L3 | 5 | 2 |
| b) | Write a neat sketch of a reciprocating pump and explain its working mechanism. | 4 | L1 | 5 | 1 |
| c) | List any four pipe fittings used in bioprocess industries. | 4 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome .
